

Supporting Information for:
An Interpretable Multilevel Model of Isocratic
Retention Explaining Between-Analyte and
Between-Column Variability

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1 Table S1. Summary of the MCMC simulations of the marginal posterior distributions of population-level model parameters.

Summary of posterior parameter estimates. The table reports posterior summaries for each parameter, including measures of central tendency (mean, median), uncertainty (standard deviation, median absolute deviation, and 5th/95th percentiles), and MCMC diagnostic statistics (rhat, bulk effective sample size, and tail effective sample size).

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
logkw_hat	3.1600	3.160	0.149	0.142	2.91000	3.4000	1.01	465	1200
S1_hat	3.9500	3.950	0.124	0.119	3.74000	4.1500	1.01	422	841
logS2_hat	0.3230	0.323	0.032	0.033	0.27000	0.3760	1.00	1050	2210
fid_hat	0.0121	0.015	0.009	0.007	-0.00670	0.0212	1.00	1640	2490
alpha_hat	4.7300	4.730	0.062	0.063	4.63000	4.8400	1.00	3130	3520
dlogkw_hat	-2.3800	-2.380	0.083	0.081	-2.51000	-2.2500	1.00	2050	2610
dS1_hat	2.1600	2.160	0.251	0.245	1.75000	2.5700	1.00	3710	2780
dlogS2_hat	-0.8620	-0.862	0.054	0.053	-0.95200	-0.7720	1.00	1850	2760
gamma_hat	3.0600	3.090	0.412	0.416	2.36000	3.7100	1.00	2730	2820
beta_a_hat[1]	0.8310	0.831	0.055	0.054	0.74000	0.9220	1.01	468	697
beta_a_hat[2]	0.7400	0.740	0.054	0.053	0.65200	0.8280	1.01	554	739
beta_c_hat[1]	0.7360	0.737	0.221	0.211	0.37000	1.1000	1.00	4890	2820
beta_c_hat[2]	0.5430	0.547	0.185	0.184	0.24000	0.8510	1.00	3930	2780
beta_c_hat[3]	0.2810	0.277	0.095	0.091	0.12800	0.4420	1.00	2640	2480
omega_a[1]	0.9920	0.988	0.071	0.072	0.88300	1.1100	1.01	698	1340
omega_a[2]	0.9160	0.914	0.069	0.069	0.80700	1.0300	1.01	709	1680
omega_a[3]	0.3450	0.344	0.025	0.024	0.30600	0.3880	1.01	723	1390
omega_c[1]	0.4260	0.415	0.088	0.081	0.30700	0.5870	1.00	2540	2550
omega_c[2]	0.3010	0.293	0.061	0.057	0.21700	0.4140	1.00	2380	2820
omega_c[3]	0.1010	0.098	0.021	0.019	0.07200	0.1380	1.00	2690	3030
omega_c[4]	1.2600	1.240	0.272	0.274	0.85300	1.7400	1.00	3520	3080
omega_c[5]	0.0119	0.009	0.008	0.006	0.00371	0.0284	1.00	1950	3040
omega_c[6]	0.2110	0.207	0.039	0.037	0.15300	0.2800	1.00	3160	2750
omega_c[7]	0.4660	0.465	0.069	0.069	0.35300	0.5820	1.00	3650	2300
omega_c[8]	0.1620	0.158	0.033	0.032	0.11600	0.2210	1.00	3870	3240
m_logomega_d[1]	-1.7900	-1.790	0.199	0.190	-2.10000	-1.4400	1.01	1390	1650
m_logomega_d[2]	-2.0100	-2.010	0.184	0.178	-2.30000	-1.7000	1.00	2230	2270
m_logomega_d[3]	-2.7100	-2.730	0.400	0.384	-3.34000	-2.0300	1.00	2060	2700
sd_logomega_d[1]	0.7130	0.692	0.143	0.132	0.52200	0.9730	1.00	1990	2720
sd_logomega_d[2]	0.6350	0.614	0.131	0.119	0.45700	0.8760	1.00	2930	2670
sd_logomega_d[3]	1.4400	1.400	0.306	0.282	1.02000	1.9900	1.00	1720	2660
rho_a[1,1]	1.0000	1.000	0.000	0.000	1.00000	1.0000	NA	NA	NA
rho_a[2,1]	0.9380	0.940	0.014	0.013	0.91200	0.9580	1.01	885	1800
rho_a[3,1]	0.4450	0.450	0.077	0.078	0.30900	0.5630	1.01	798	1290
rho_a[1,2]	0.9380	0.940	0.014	0.013	0.91200	0.9580	1.01	885	1800
rho_a[2,2]	1.0000	1.000	0.000	0.000	1.00000	1.0000	NA	NA	NA
rho_a[3,2]	0.3510	0.355	0.084	0.085	0.20500	0.4840	1.01	646	1460
rho_a[1,3]	0.4450	0.450	0.077	0.078	0.30900	0.5630	1.01	798	1290
rho_a[2,3]	0.3510	0.355	0.084	0.085	0.20500	0.4840	1.01	646	1460
rho_a[3,3]	1.0000	1.000	0.000	0.000	1.00000	1.0000	NA	NA	NA
rho_c[1,1]	1.0000	1.000	0.000	0.000	1.00000	1.0000	NA	NA	NA
rho_c[2,1]	0.8290	0.847	0.094	0.087	0.65300	0.9460	1.00	2750	2550
rho_c[3,1]	0.7090	0.727	0.123	0.122	0.47700	0.8820	1.00	3610	2760

(continued)

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
rho_c[1,2]	0.8290	0.847	0.094	0.087	0.65300	0.9460	1.00	2750	2550
rho_c[2,2]	1.0000	1.000	0.000	0.000	1.00000	1.0000	NA	NA	NA
rho_c[3,2]	0.7440	0.763	0.116	0.112	0.53100	0.8990	1.00	3100	2550
rho_c[1,3]	0.7090	0.727	0.123	0.122	0.47700	0.8820	1.00	3610	2760
rho_c[2,3]	0.7440	0.763	0.116	0.112	0.53100	0.8990	1.00	3100	2550
rho_c[3,3]	1.0000	1.000	0.000	0.000	1.00000	1.0000	NA	NA	NA
rho_d[1,1]	1.0000	1.000	0.000	0.000	1.00000	1.0000	NA	NA	NA
rho_d[2,1]	0.4300	0.444	0.190	0.191	0.08560	0.7140	1.00	3570	3000
rho_d[3,1]	0.5950	0.618	0.173	0.170	0.27500	0.8340	1.00	3540	2950
rho_d[1,2]	0.4300	0.444	0.190	0.191	0.08560	0.7140	1.00	3570	3000
rho_d[2,2]	1.0000	1.000	0.000	0.000	1.00000	1.0000	NA	NA	NA
rho_d[3,2]	0.4460	0.467	0.184	0.190	0.12200	0.7110	1.00	3750	3350
rho_d[1,3]	0.5950	0.618	0.173	0.170	0.27500	0.8340	1.00	3540	2950
rho_d[2,3]	0.4460	0.467	0.184	0.190	0.12200	0.7110	1.00	3750	3350
rho_d[3,3]	1.0000	1.000	0.000	0.000	1.00000	1.0000	NA	NA	NA
sigma	0.0207	0.021	0.000	0.000	0.02040	0.0211	1.00	1460	2580

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
kappa[1,1]	-0.529000	-0.528	0.150	0.146	-0.78000	-0.290000	1.00	1830	2320
kappa[2,1]	-1.070000	-1.070	0.138	0.137	-1.30000	-0.849000	1.00	1410	1920
kappa[3,1]	-0.045200	-0.045	0.123	0.117	-0.24300	0.156000	1.00	1440	1660
kappa[4,1]	0.158000	0.158	0.123	0.117	-0.03790	0.362000	1.00	1600	1800
kappa[5,1]	0.080400	0.077	0.119	0.114	-0.10700	0.281000	1.00	1260	1980
kappa[6,1]	0.080300	0.077	0.126	0.120	-0.12100	0.291000	1.00	1520	2020
kappa[7,1]	-0.006750	-0.008	0.128	0.122	-0.21200	0.208000	1.00	1520	2190
kappa[8,1]	0.179000	0.179	0.118	0.113	-0.00791	0.375000	1.00	1330	1780
kappa[9,1]	0.321000	0.319	0.120	0.114	0.12800	0.518000	1.00	1530	1880
kappa[10,1]	0.068500	0.067	0.123	0.117	-0.13000	0.274000	1.00	1440	2100
kappa[11,1]	0.268000	0.265	0.126	0.122	0.06500	0.480000	1.00	1530	2300
kappa[12,1]	0.485000	0.483	0.114	0.109	0.30500	0.675000	1.00	1390	1770
kappa[13,1]	0.467000	0.464	0.116	0.111	0.28200	0.663000	1.00	1420	1920
kappa[1,2]	-0.082200	-0.083	0.111	0.107	-0.26400	0.102000	1.00	2550	2710
kappa[2,2]	-0.849000	-0.848	0.105	0.101	-1.03000	-0.679000	1.00	1680	2250
kappa[3,2]	-0.137000	-0.138	0.087	0.084	-0.27900	0.007770	1.00	1760	1980
kappa[4,2]	0.174000	0.174	0.089	0.085	0.02980	0.322000	1.00	1870	2380
kappa[5,2]	-0.093600	-0.096	0.087	0.086	-0.23400	0.049100	1.00	1490	2210
kappa[6,2]	0.061500	0.059	0.092	0.090	-0.08160	0.217000	1.00	1810	2170
kappa[7,2]	0.012600	0.011	0.094	0.093	-0.13700	0.165000	1.00	1950	1960
kappa[8,2]	0.065200	0.064	0.083	0.082	-0.06820	0.200000	1.00	1720	2240
kappa[9,2]	0.204000	0.203	0.086	0.081	0.06570	0.343000	1.00	1850	2210
kappa[10,2]	0.134000	0.131	0.090	0.090	-0.00518	0.285000	1.00	1790	1900
kappa[11,2]	0.278000	0.275	0.093	0.093	0.13100	0.435000	1.00	1840	2150
kappa[12,2]	0.280000	0.278	0.082	0.080	0.15200	0.415000	1.00	1670	1830
kappa[13,2]	0.251000	0.250	0.083	0.082	0.12200	0.388000	1.00	1690	1770
kappa[1,3]	-0.076500	-0.077	0.056	0.056	-0.17000	0.014500	1.00	1900	2190
kappa[2,3]	-0.252000	-0.250	0.051	0.051	-0.33700	-0.170000	1.00	2010	2380
kappa[3,3]	-0.042900	-0.043	0.036	0.035	-0.10100	0.016600	1.00	1770	2560
kappa[4,3]	0.152000	0.152	0.034	0.033	0.09680	0.210000	1.01	2140	2290
kappa[5,3]	0.000298	0.000	0.031	0.030	-0.04800	0.051800	1.00	1730	1840
kappa[6,3]	0.012400	0.013	0.033	0.032	-0.04010	0.067200	1.00	1780	2220
kappa[7,3]	0.063100	0.062	0.036	0.035	0.00554	0.124000	1.00	1810	2410
kappa[8,3]	0.017000	0.017	0.029	0.029	-0.03150	0.066200	1.00	1720	2130
kappa[9,3]	0.107000	0.107	0.032	0.030	0.05470	0.160000	1.01	2040	2080
kappa[10,3]	0.043000	0.043	0.031	0.032	-0.00714	0.096800	1.00	1700	1860
kappa[11,3]	0.005400	0.005	0.033	0.033	-0.04860	0.061700	1.00	1810	2340
kappa[12,3]	0.057300	0.056	0.026	0.025	0.01530	0.101000	1.00	1590	1950
kappa[13,3]	0.032400	0.031	0.027	0.026	-0.01060	0.077700	1.00	1620	2200
kappa[1,4]	0.600000	0.569	0.621	0.611	-0.37300	1.670000	1.00	3890	3320
kappa[2,4]	1.100000	1.080	0.648	0.655	0.07260	2.210000	1.00	3100	2940
kappa[3,4]	0.578000	0.550	0.625	0.622	-0.39400	1.650000	1.00	3160	3050
kappa[4,4]	1.340000	1.310	0.571	0.569	0.43700	2.320000	1.00	3370	3140
kappa[5,4]	-0.529000	-0.554	0.570	0.568	-1.43000	0.427000	1.00	3730	3160
kappa[6,4]	0.433000	0.411	0.526	0.528	-0.39500	1.320000	1.00	3590	3410
kappa[7,4]	1.610000	1.590	0.597	0.593	0.69700	2.650000	1.00	2930	2870
kappa[8,4]	3.530000	3.500	0.636	0.647	2.53000	4.630000	1.00	2760	2660
kappa[9,4]	0.554000	0.534	0.513	0.502	-0.22900	1.440000	1.00	3120	3110
kappa[10,4]	-0.375000	-0.398	0.544	0.543	-1.24000	0.534000	1.00	3160	2370
kappa[11,4]	1.440000	1.400	0.596	0.591	0.49800	2.460000	1.00	2550	2240
kappa[12,4]	1.260000	1.220	0.608	0.613	0.31400	2.310000	1.00	3190	2810
kappa[13,4]	1.430000	1.400	0.521	0.510	0.62400	2.340000	1.00	3140	2490
kappa[1,5]	0.014700	0.012	0.009	0.007	0.00528	0.033700	1.00	1670	2370
kappa[2,5]	-0.004310	-0.003	0.012	0.007	-0.02640	0.010500	1.00	4250	3190
kappa[3,5]	-0.002880	-0.001	0.012	0.008	-0.02290	0.012200	1.00	4130	3220
kappa[4,5]	-0.003770	-0.002	0.011	0.006	-0.02300	0.010700	1.00	3340	3360
kappa[5,5]	0.013600	0.011	0.009	0.007	0.00419	0.032300	1.00	1650	2450

(continued)

variable	mean	median	sd	mad	q5	q95	rhat	ess	bulk	ess_tail
kappa[6,5]	-0.005100	-0.003	0.012	0.007	-0.02500	0.009190	1.00		3710	3110
kappa[7,5]	0.011000	0.008	0.010	0.007	0.00178	0.029600	1.01		1410	1080
kappa[8,5]	-0.004970	-0.003	0.012	0.007	-0.02620	0.009300	1.00		3520	3420
kappa[9,5]	-0.004380	-0.002	0.012	0.007	-0.02480	0.010600	1.00		3990	3030
kappa[10,5]	-0.005330	-0.003	0.012	0.007	-0.02570	0.008740	1.00		3710	3350
kappa[11,5]	-0.000662	0.001	0.012	0.007	-0.02190	0.015000	1.00		2530	2830
kappa[12,5]	-0.004130	-0.002	0.012	0.007	-0.02480	0.010300	1.00		4040	3220
kappa[13,5]	-0.004320	-0.003	0.012	0.006	-0.02280	0.009670	1.00		2930	3330
kappa[1,6]	0.324000	0.324	0.096	0.093	0.17000	0.488000	1.00		1770	2300
kappa[2,6]	0.235000	0.235	0.120	0.119	0.03720	0.428000	1.00		1740	2280
kappa[3,6]	0.420000	0.422	0.092	0.089	0.26800	0.568000	1.00		2050	2460
kappa[4,6]	-0.137000	-0.136	0.084	0.080	-0.28000	-0.000857	1.00		2260	2810
kappa[5,6]	-0.155000	-0.153	0.103	0.103	-0.32600	0.012700	1.00		1940	2210
kappa[6,6]	-0.055900	-0.054	0.072	0.069	-0.17700	0.060200	1.00		2310	2660
kappa[7,6]	-0.155000	-0.154	0.092	0.091	-0.30400	-0.006650	1.00		1400	2580
kappa[8,6]	-0.302000	-0.299	0.101	0.099	-0.47200	-0.137000	1.00		2330	2770
kappa[9,6]	0.059400	0.060	0.078	0.075	-0.06730	0.184000	1.00		2360	2580
kappa[10,6]	0.001980	0.003	0.077	0.075	-0.12400	0.125000	1.00		2290	2450
kappa[11,6]	-0.272000	-0.268	0.087	0.087	-0.42200	-0.136000	1.00		2390	3060
kappa[12,6]	-0.055100	-0.054	0.073	0.069	-0.17700	0.061900	1.00		2340	2700
kappa[13,6]	-0.159000	-0.158	0.073	0.069	-0.27900	-0.042900	1.00		2290	2600
kappa[1,7]	0.076000	0.073	0.393	0.380	-0.55700	0.729000	1.00		6270	3400
kappa[2,7]	-0.402000	-0.393	0.423	0.423	-1.11000	0.274000	1.00		5520	2760
kappa[3,7]	-0.447000	-0.443	0.381	0.382	-1.07000	0.169000	1.00		4530	3240
kappa[4,7]	-0.448000	-0.450	0.328	0.320	-0.98000	0.088800	1.00		5250	3300
kappa[5,7]	0.608000	0.601	0.405	0.406	-0.03850	1.300000	1.00		6420	3500
kappa[6,7]	-0.034400	-0.041	0.317	0.315	-0.54200	0.499000	1.00		6430	3310
kappa[7,7]	1.100000	1.080	0.377	0.364	0.50300	1.740000	1.00		3710	3290
kappa[8,7]	1.280000	1.270	0.392	0.385	0.66700	1.950000	1.00		3580	3180
kappa[9,7]	-0.352000	-0.354	0.287	0.279	-0.82200	0.124000	1.00		4340	2900
kappa[10,7]	-0.957000	-0.958	0.330	0.336	-1.49000	-0.419000	1.00		4090	3080
kappa[11,7]	0.226000	0.216	0.349	0.344	-0.32600	0.816000	1.00		3680	3130
kappa[12,7]	0.951000	0.932	0.388	0.384	0.34500	1.630000	1.00		4530	3100
kappa[13,7]	0.297000	0.288	0.304	0.305	-0.17500	0.803000	1.00		5900	3270
kappa[1,8]	0.028200	0.028	0.079	0.078	-0.10000	0.158000	1.00		2000	2790
kappa[2,8]	0.174000	0.173	0.079	0.080	0.04870	0.304000	1.00		2450	3090
kappa[3,8]	0.262000	0.261	0.068	0.067	0.15200	0.372000	1.00		1960	2790
kappa[4,8]	0.071500	0.072	0.057	0.054	-0.02230	0.166000	1.00		2540	2840
kappa[5,8]	-0.323000	-0.321	0.073	0.072	-0.44400	-0.206000	1.00		2370	3080
kappa[6,8]	-0.015000	-0.016	0.051	0.050	-0.09950	0.069300	1.00		2070	2500
kappa[7,8]	-0.272000	-0.269	0.072	0.070	-0.39500	-0.157000	1.00		1750	2580
kappa[8,8]	-0.096400	-0.096	0.063	0.061	-0.20100	0.005800	1.00		2120	2350
kappa[9,8]	0.080700	0.081	0.052	0.050	-0.00347	0.166000	1.00		2110	2680
kappa[10,8]	0.064500	0.064	0.054	0.053	-0.02220	0.152000	1.00		2020	2830
kappa[11,8]	0.014700	0.015	0.054	0.053	-0.07540	0.104000	1.00		1950	2760
kappa[12,8]	-0.064800	-0.065	0.051	0.048	-0.14700	0.019300	1.00		2070	2650
kappa[13,8]	-0.007820	-0.007	0.051	0.049	-0.09190	0.075200	1.00		2110	2520

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
omega_d1[1]	0.32300	0.321	0.028	0.028	0.280000	0.3720	1.00	1780	2570
omega_d1[2]	0.51800	0.516	0.043	0.043	0.450000	0.5890	1.00	1260	1880
omega_d1[3]	0.22000	0.218	0.020	0.021	0.189000	0.2550	1.00	2250	2780
omega_d1[4]	0.12900	0.128	0.014	0.013	0.108000	0.1530	1.00	2830	2640
omega_d1[5]	0.28200	0.280	0.029	0.029	0.237000	0.3330	1.01	1760	2520
omega_d1[6]	0.05320	0.053	0.007	0.007	0.041400	0.0656	1.00	1580	2460
omega_d1[7]	0.17300	0.172	0.021	0.020	0.141000	0.2090	1.00	1210	1700
omega_d1[8]	0.28100	0.279	0.027	0.026	0.240000	0.3270	1.00	1990	2580
omega_d1[9]	0.12300	0.123	0.013	0.013	0.103000	0.1460	1.00	1820	2340
omega_d1[10]	0.07620	0.076	0.009	0.009	0.062500	0.0910	1.00	2840	2790
omega_d1[11]	0.15500	0.154	0.014	0.014	0.133000	0.1790	1.00	2300	3330
omega_d1[12]	0.05360	0.053	0.009	0.008	0.040300	0.0694	1.00	1170	2240
omega_d1[13]	0.04130	0.041	0.009	0.009	0.027600	0.0561	1.01	931	1980
omega_d2[1]	0.20100	0.200	0.021	0.020	0.169000	0.2370	1.00	2790	3090
omega_d2[2]	0.43100	0.429	0.037	0.038	0.373000	0.4950	1.00	1810	2270
omega_d2[3]	0.11600	0.115	0.016	0.016	0.091700	0.1430	1.00	2430	3050
omega_d2[4]	0.16300	0.162	0.016	0.016	0.139000	0.1900	1.00	2740	2990
omega_d2[5]	0.28800	0.287	0.028	0.027	0.246000	0.3370	1.00	2310	2970
omega_d2[6]	0.05410	0.054	0.009	0.009	0.039400	0.0695	1.00	1480	2190
omega_d2[7]	0.17000	0.169	0.021	0.021	0.137000	0.2060	1.00	961	1810
omega_d2[8]	0.12600	0.124	0.017	0.017	0.099100	0.1560	1.00	1900	2840
omega_d2[9]	0.04890	0.048	0.011	0.010	0.031900	0.0671	1.00	1410	2130
omega_d2[10]	0.09130	0.091	0.012	0.012	0.072900	0.1110	1.00	1990	2730
omega_d2[11]	0.09030	0.089	0.015	0.015	0.067500	0.1160	1.00	1430	2500
omega_d2[12]	0.07680	0.076	0.011	0.011	0.059700	0.0957	1.00	2070	3320
omega_d2[13]	0.06730	0.067	0.010	0.010	0.051100	0.0844	1.00	1970	2880
omega_d3[1]	0.40900	0.406	0.050	0.051	0.336000	0.4970	1.01	1260	2750
omega_d3[2]	0.32100	0.318	0.043	0.043	0.256000	0.3970	1.00	1840	2730
omega_d3[3]	0.16400	0.162	0.020	0.020	0.132000	0.2000	1.01	1680	2610
omega_d3[4]	0.05090	0.051	0.007	0.007	0.040100	0.0628	1.00	2530	2950
omega_d3[5]	0.15900	0.158	0.017	0.016	0.134000	0.1890	1.00	2380	3100
omega_d3[6]	0.00491	0.004	0.003	0.003	0.000921	0.0103	1.00	2530	2640
omega_d3[7]	0.15200	0.151	0.015	0.015	0.129000	0.1790	1.00	1530	2350
omega_d3[8]	0.10500	0.105	0.012	0.012	0.086000	0.1260	1.00	2530	2940
omega_d3[9]	0.01820	0.018	0.007	0.007	0.006050	0.0293	1.01	744	1210
omega_d3[10]	0.00840	0.008	0.005	0.005	0.001660	0.0169	1.00	1180	2020
omega_d3[11]	0.03630	0.036	0.008	0.008	0.023000	0.0501	1.01	1070	1730
omega_d3[12]	0.00743	0.007	0.005	0.005	0.001240	0.0168	1.00	1070	1700
omega_d3[13]	0.01110	0.011	0.005	0.005	0.002500	0.0189	1.00	941	1730

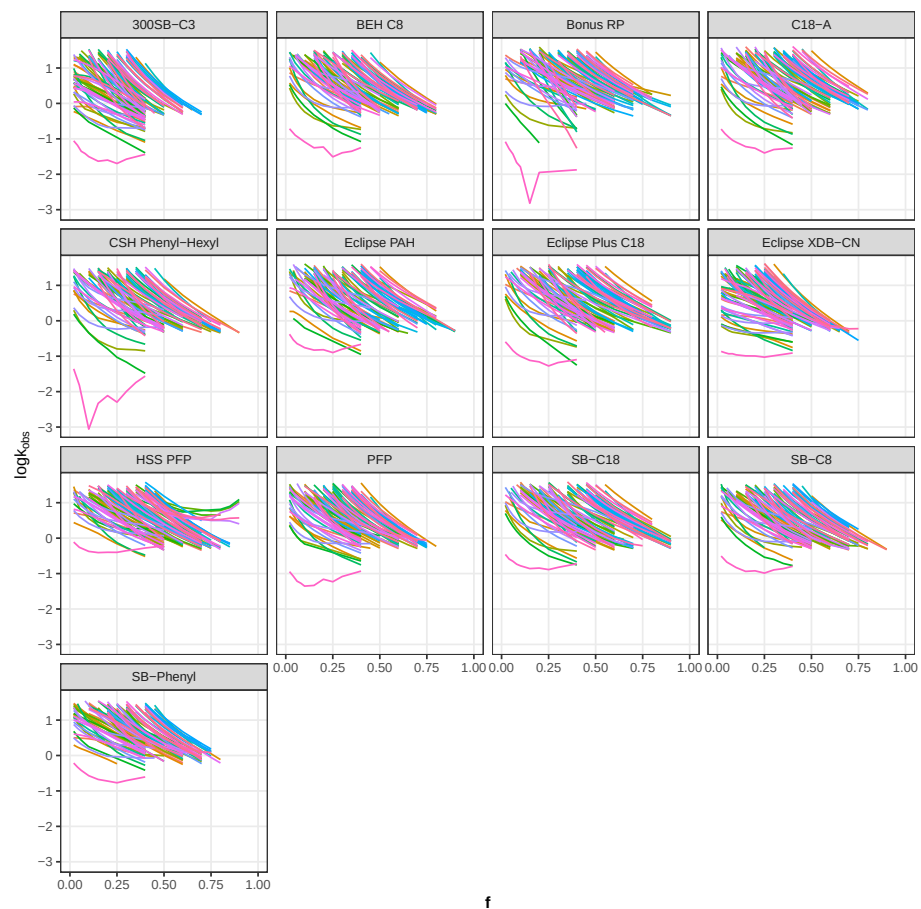
2 Table S2. Column Characteristic

No	Phase	PS	Ligand	Endcapping	CL	mm.x.mm..um	Other	Abbr.
1	Agilent 300SB-C3	300	C3	no	3.1	12.5 x 2.1, 5	Diisopropyl silane	300SB-C3
2	Agilent Eclipse XDB-CN	80	CN	double	4.3	12.5 x 2.1, 5	Dense bonding	Eclipse XDB-CN
3	Agilent SB-Phenyl	80	phenyl	no	5.5	5 x 2.1, 1.8	Diisopropyl silane	SB-Phenyl
4	Agilent PFP	120	PFP	yes	5.1	5 x 2.1, 1.9	Superficially porous	PFP
5	Agilent Bonus RP	80	C14	triple	9.5	5 x 2.1, 1.8	Diisopropyl silane Embedded amide	Bonus RP
6	Varian/Agilent C18-A	180	C18	yes	13.8	10 x 2.0, 3	Polar modified	C18-A
7	Waters CSH Phenyl-Hexyl	130	phenyl-hexyl	yes	14.0	5 x 2.1, 3.5	Low silanol activity	CSH Phenyl-Hexyl
8	Waters HSS PFP	100	PFP	no	7.0	5 x 2.1, 3.5	High silanol activity	HSS PFP
9	Agilent SB-C8	80	C8	no	5.5	5 x 2.1, 2.7	Diisopropyl silane	SB-C8
10	Waters BEH C8	130	C8	yes	13.0	5 x 2.1, 1.7	Low silanol activity	BEH C8
11	Agilent Eclipse PAH	95	C18	double	14.0	5 x 2.1, 1.8	Polymeric	Eclipse PAH
12	Agilent Eclipse Plus C18	95	C18	double	9.0	5 x 2.1, 1.8	Dense monolayer	Eclipse Plus C18
13	Agilent SB-C18	80	C18	no	10.0	5 x 2.1, 1.8	Diisobutyl silane	SB-C18

Note:

CL: carbon loading, PS: Pore Size

3 Figure S1. Raw data.



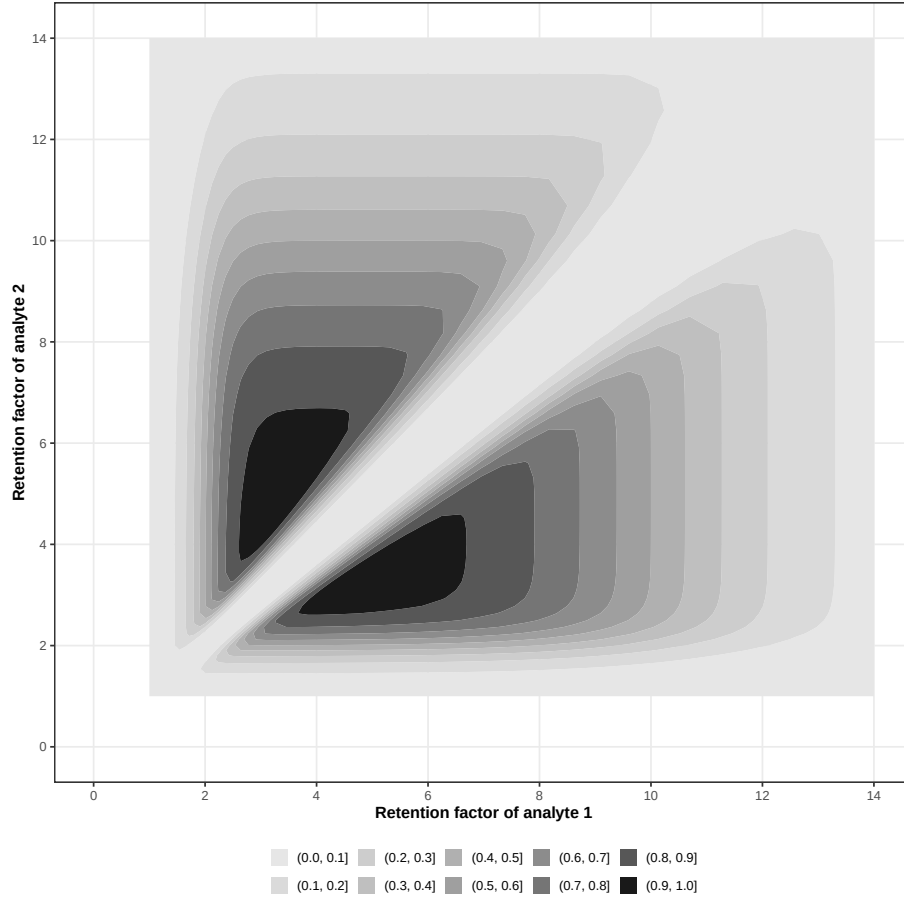
4 Figure S2. Utility function.

$$d = |\log k_1 - \log k_2|,$$

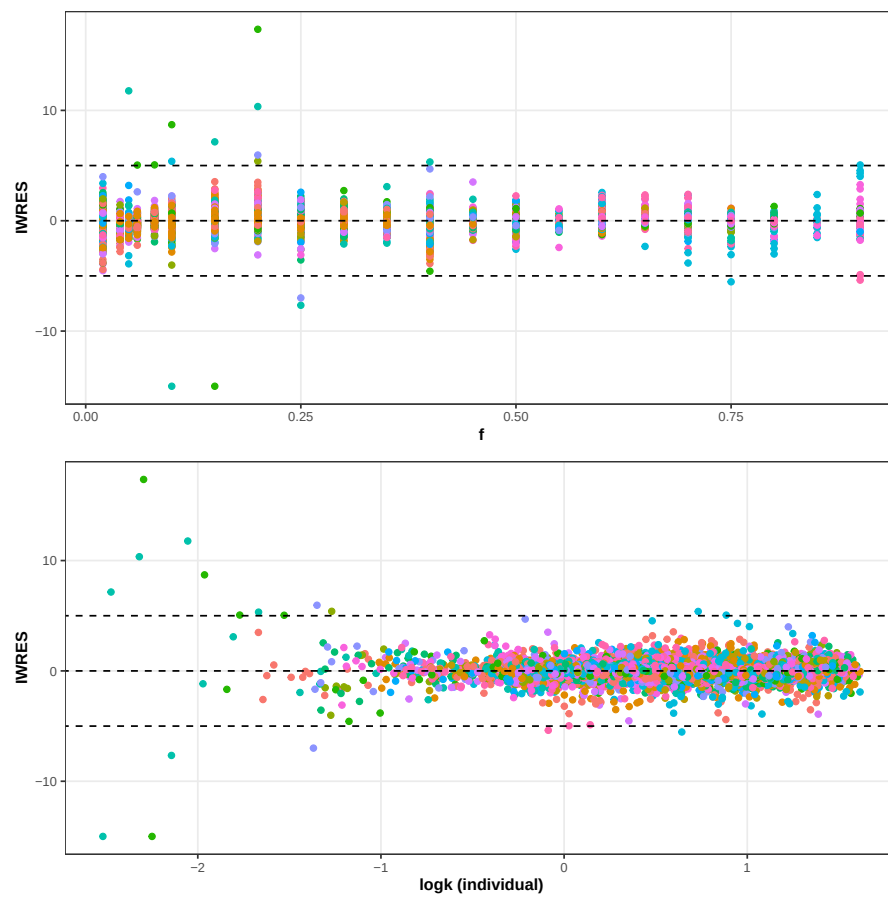
$$l_o = 10^{\min(\log k_1, \log k_2)},$$

$$h_i = 10^{\max(\log k_1, \log k_2)},$$

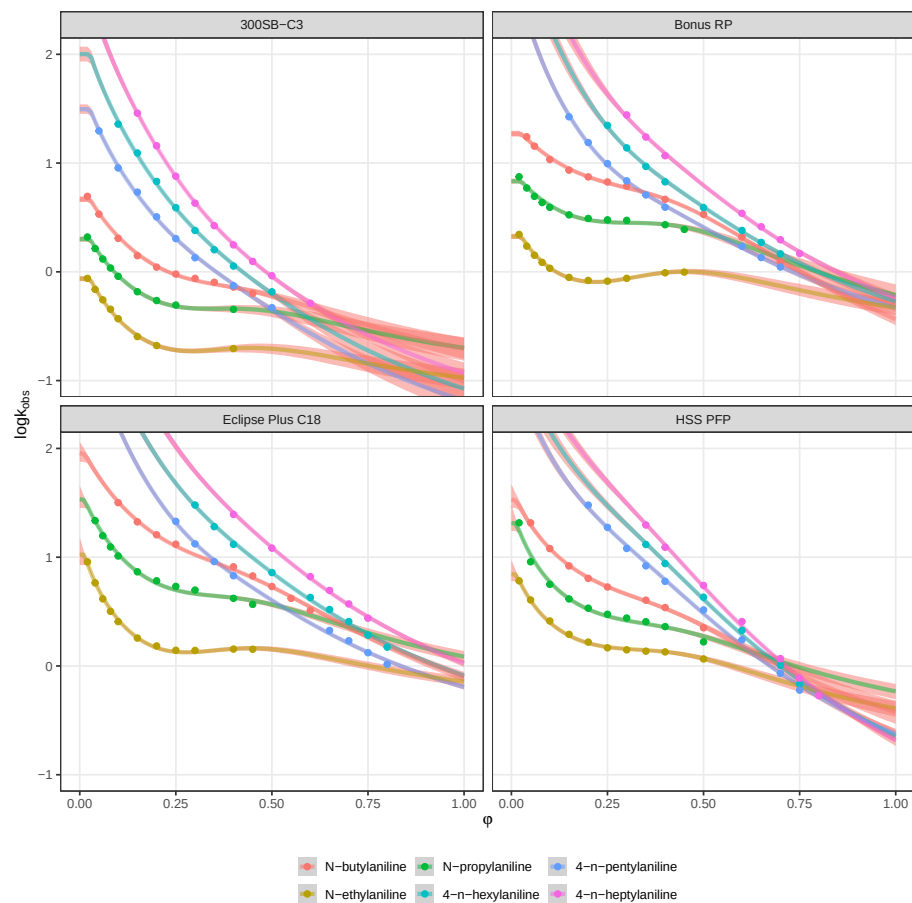
$$U = \frac{1}{1 + e^{-(d-0.07918125)/0.015}} \times \frac{1}{1 + e^{-(l_o-2)/0.25}} \times \frac{1}{1 + e^{-(10-h_i)/1.5}}.$$



5 Figure S3. Residuals

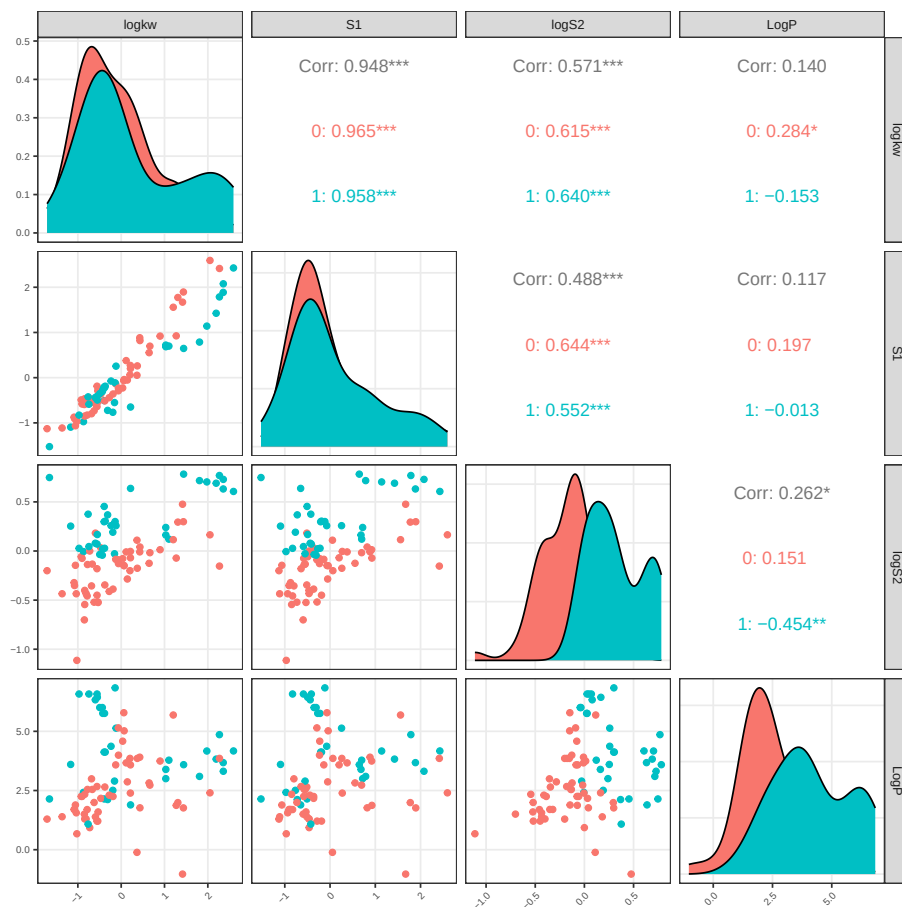


6 Figure S4. GOF plots for 6 analytes



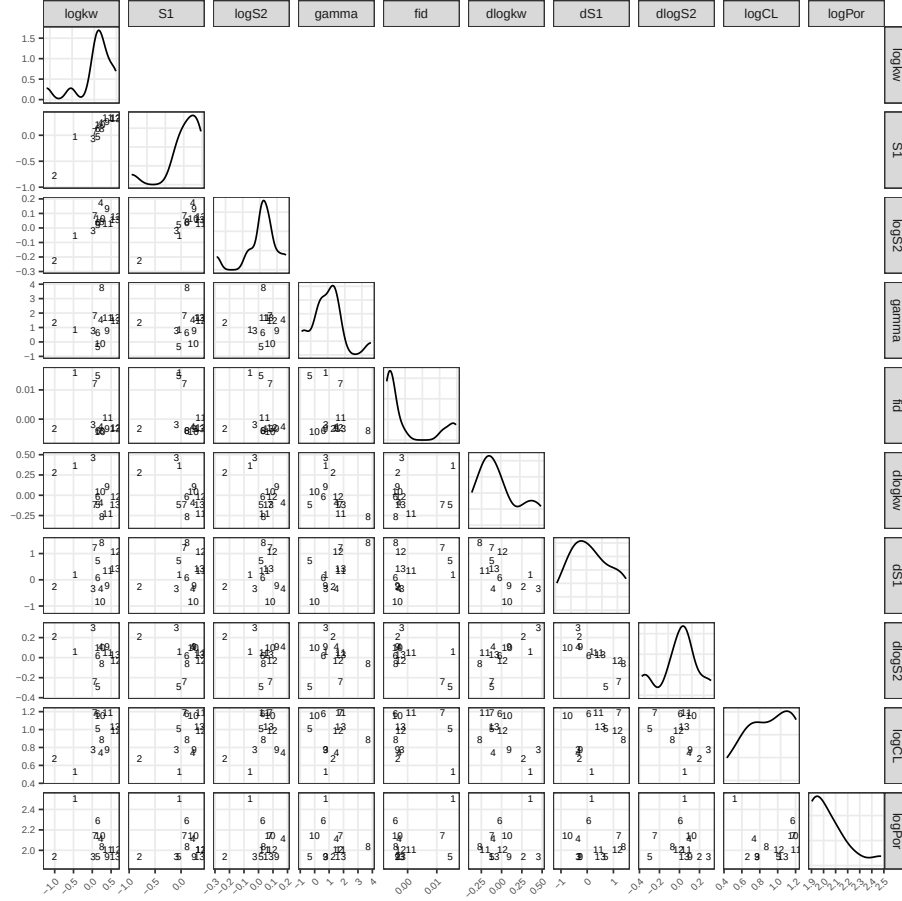
7 Figure S5. Etas

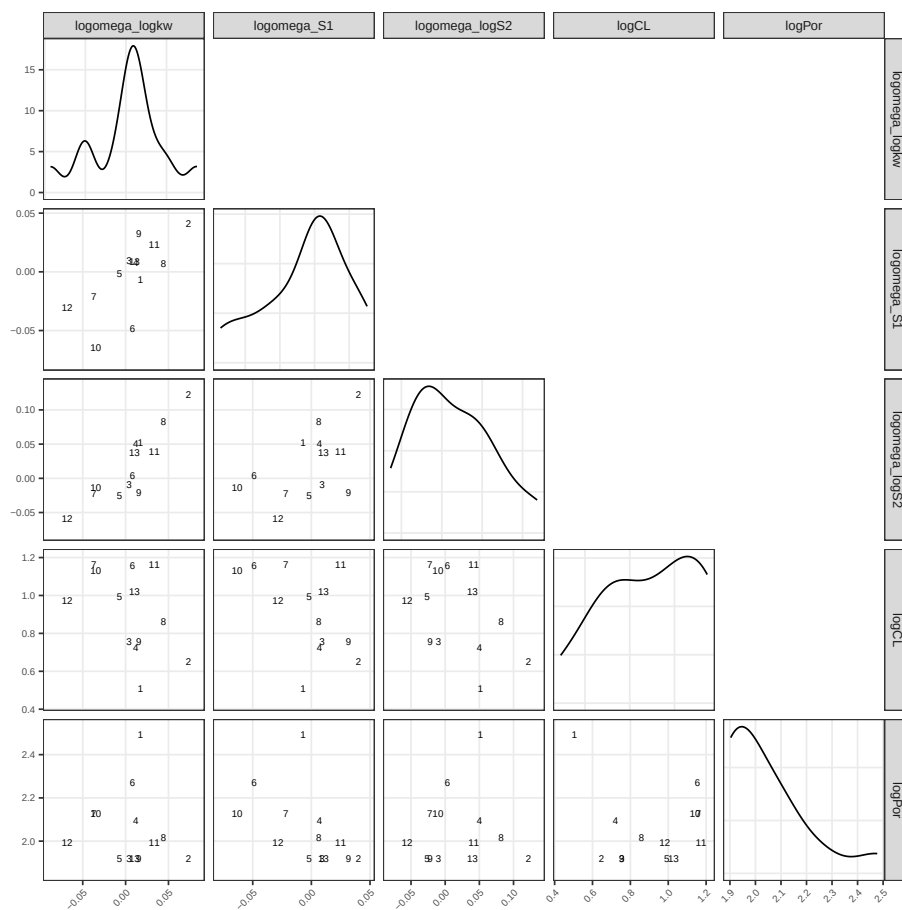
η_{a_i} represents the deviation of an individual analyte's parameter from its expected population-level value for a typical column, assuming no interactions.



8 Figure S6. Kappas

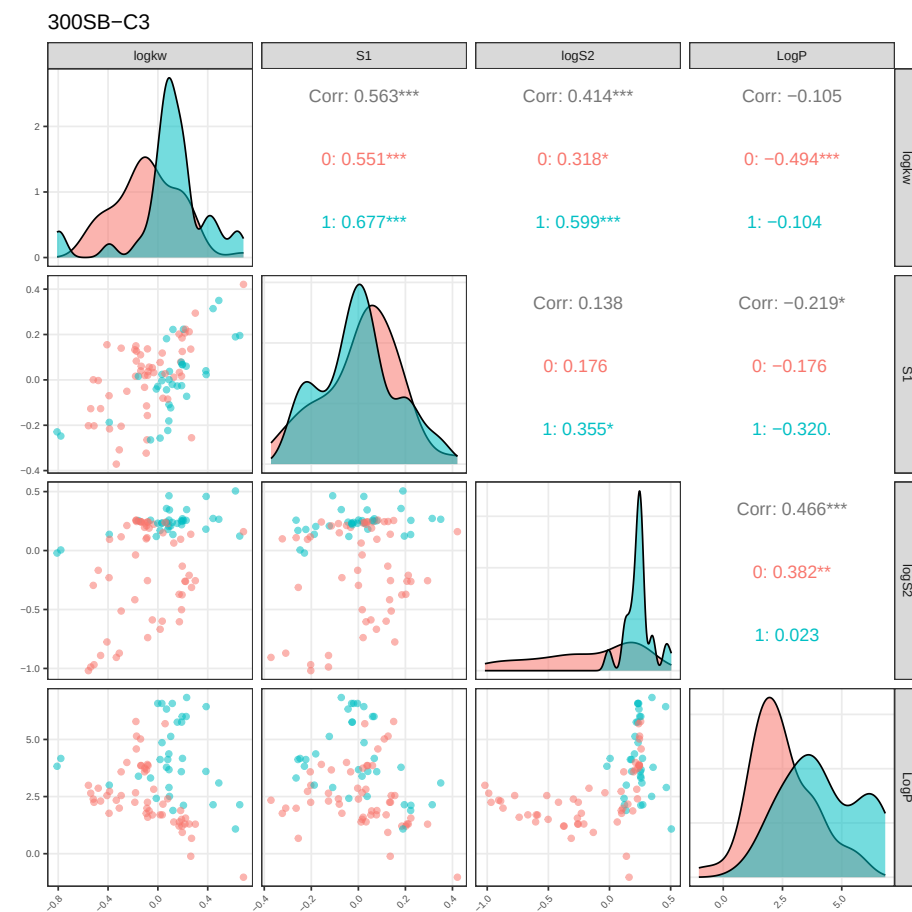
κ_c represents the deviation of an individual analyte's parameter from its expected population-level value for a typical analyte, assuming no interactions. The numbers correspond to specific columns (stationary phases). The legend is provided in Table 2S.



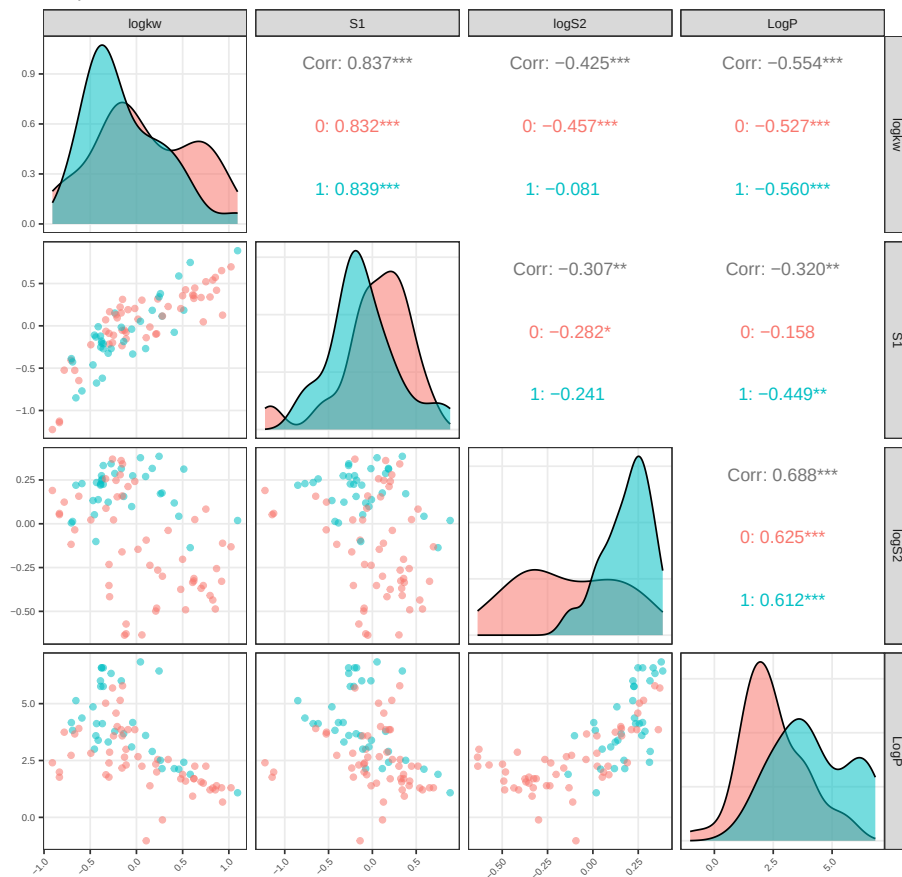


9 Figure S7. Dzetas

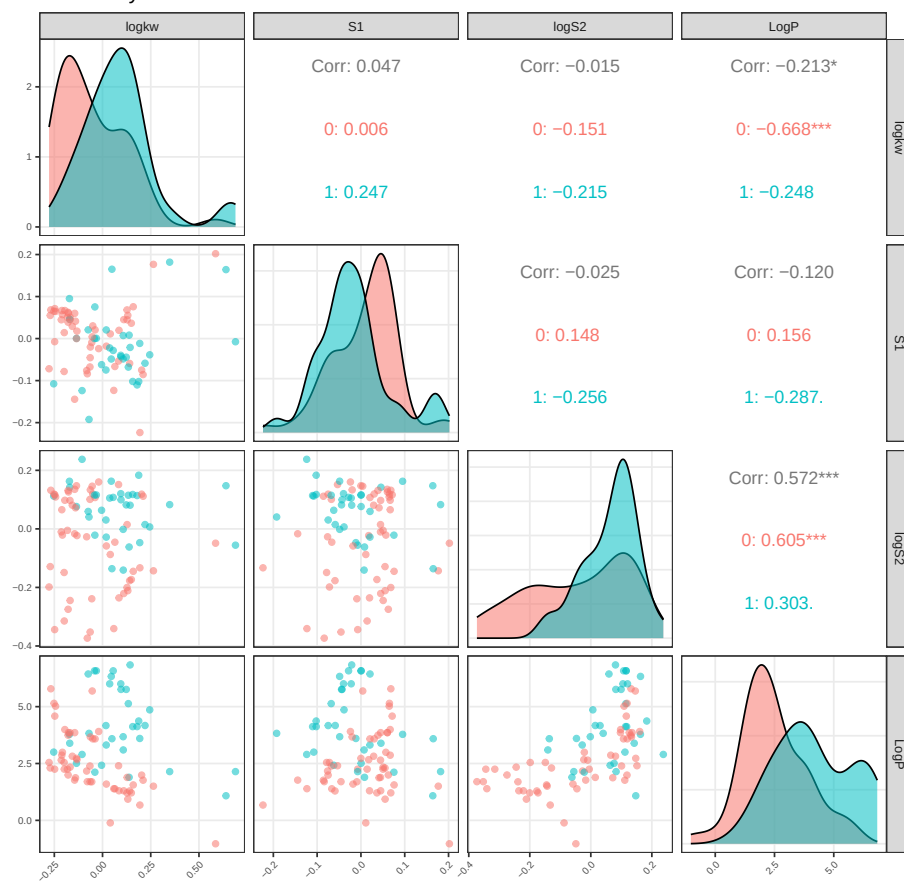
$\text{dzeta}_{a,c}$ represents analyte-column interactions.



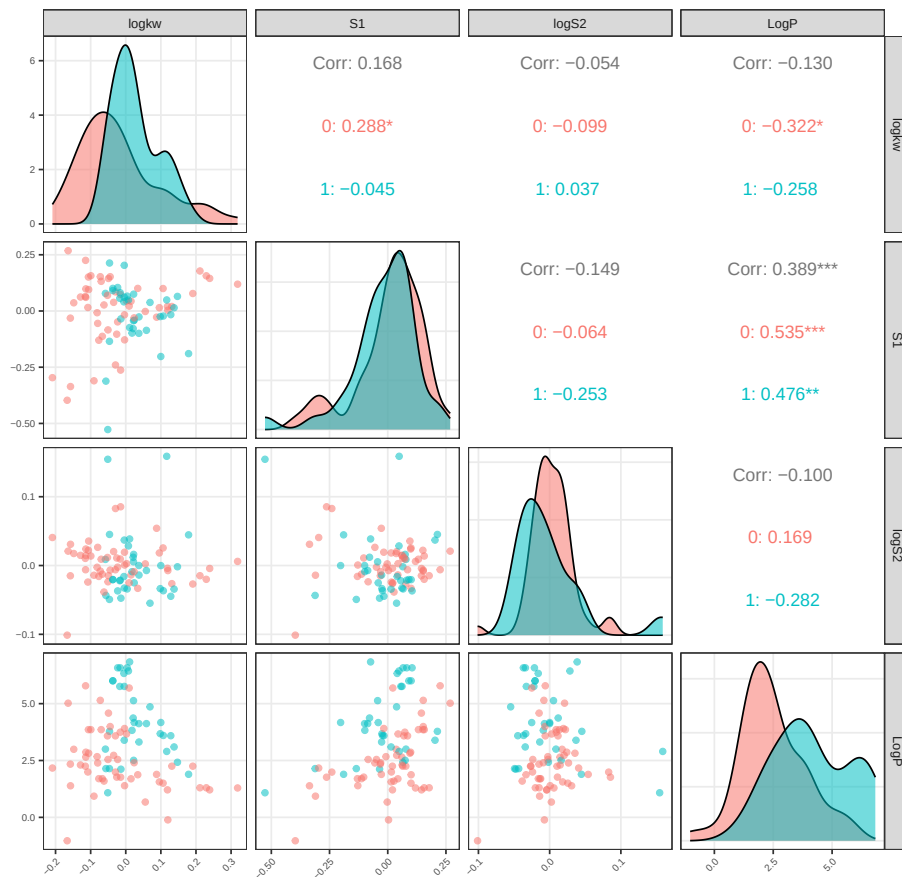
Eclipse XDB-CN



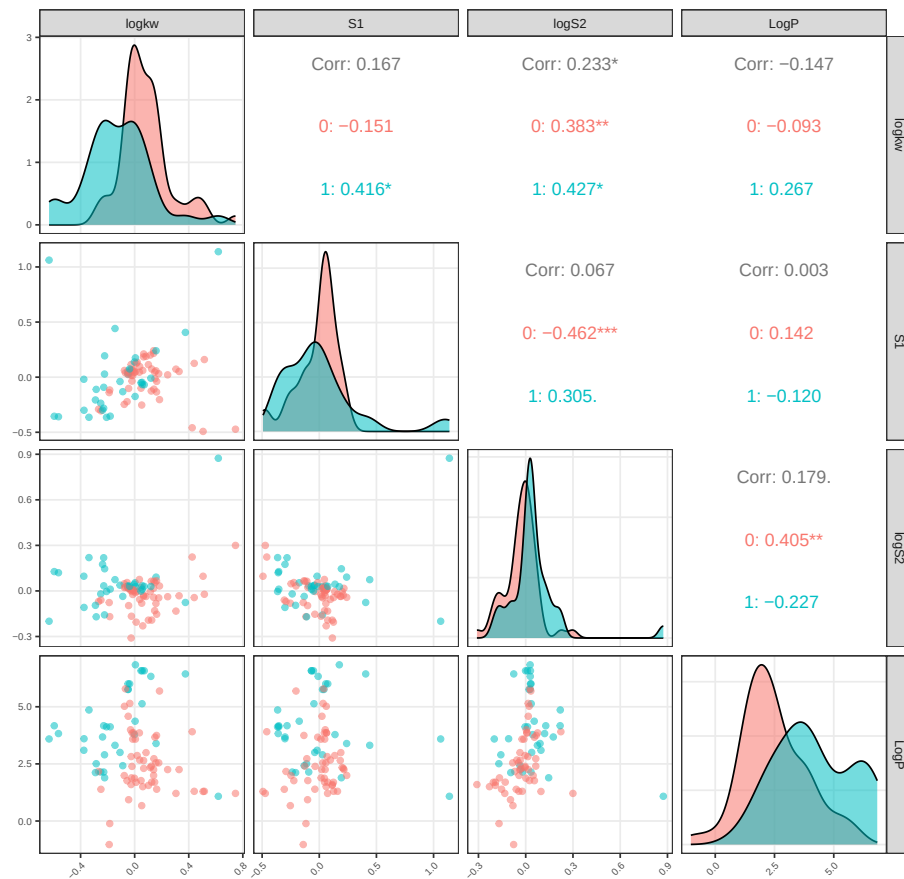
SB-Phenyl



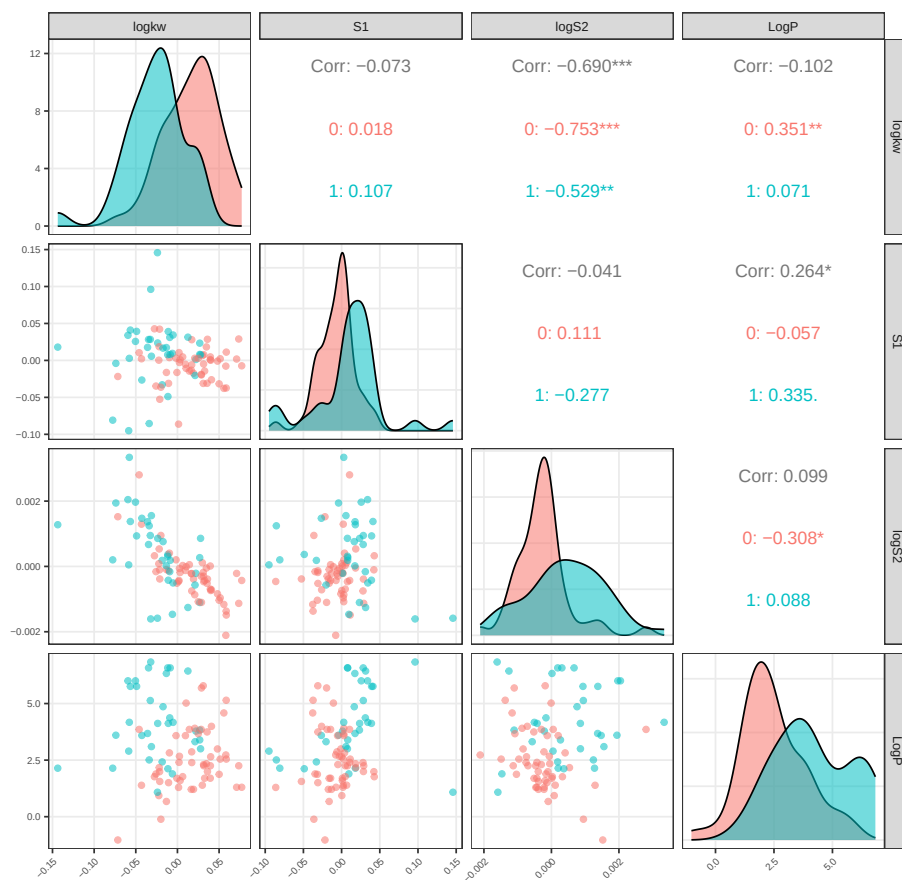
PFP



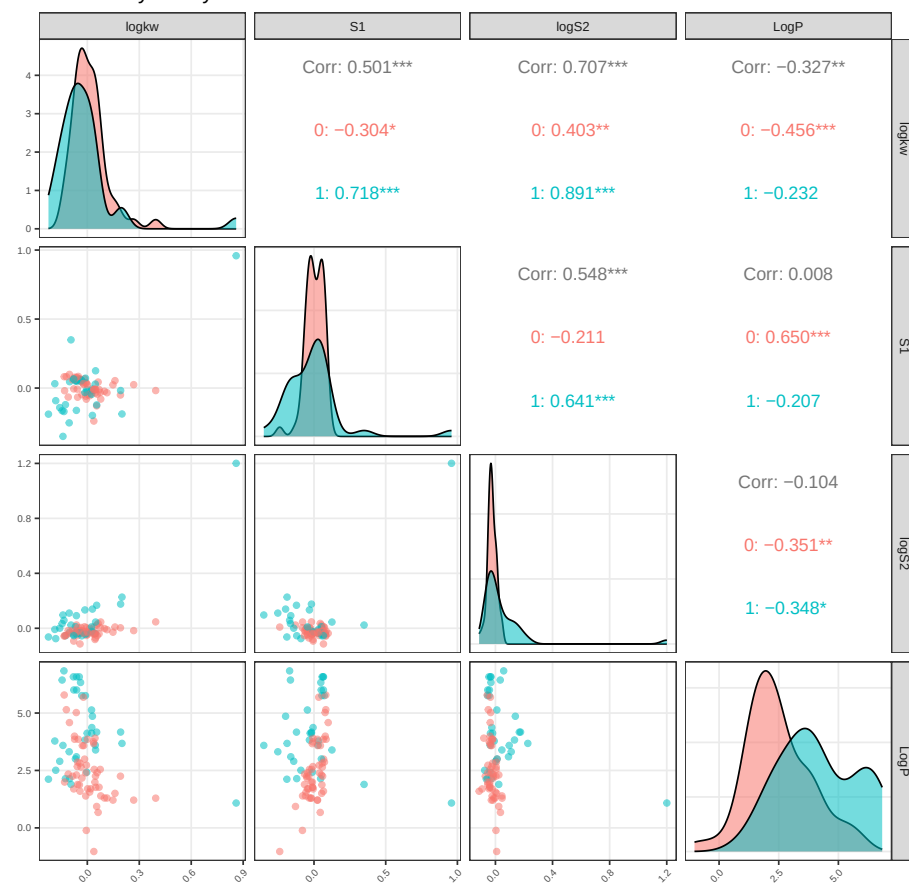
Bonus RP



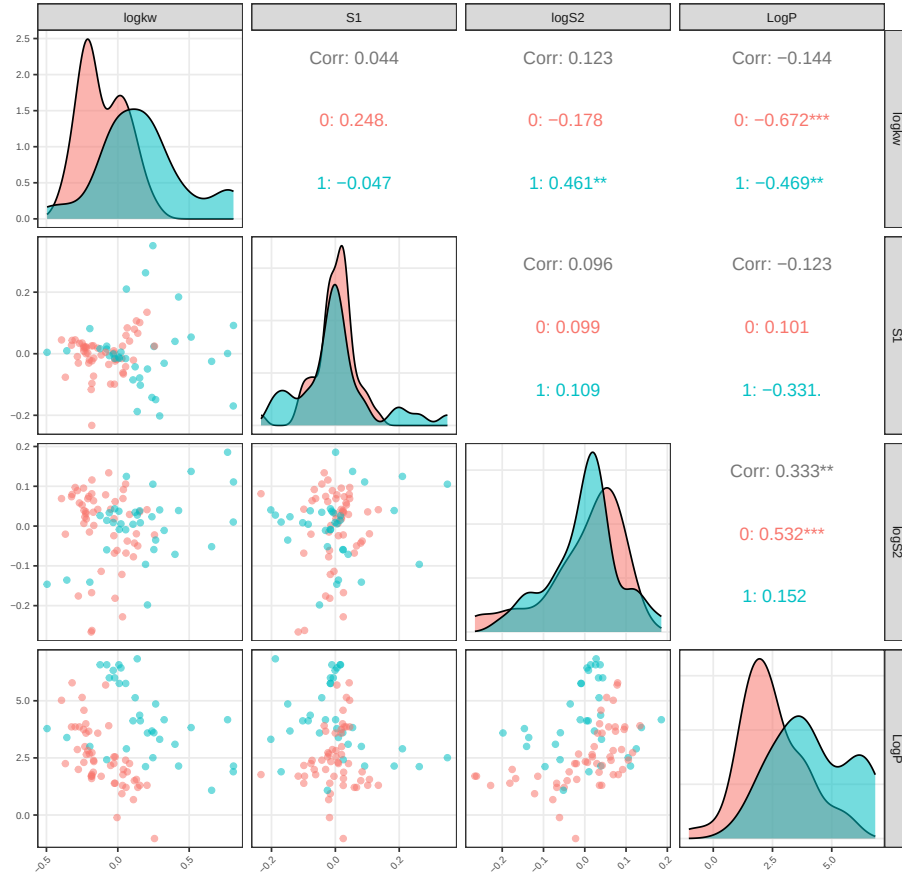
C18-A



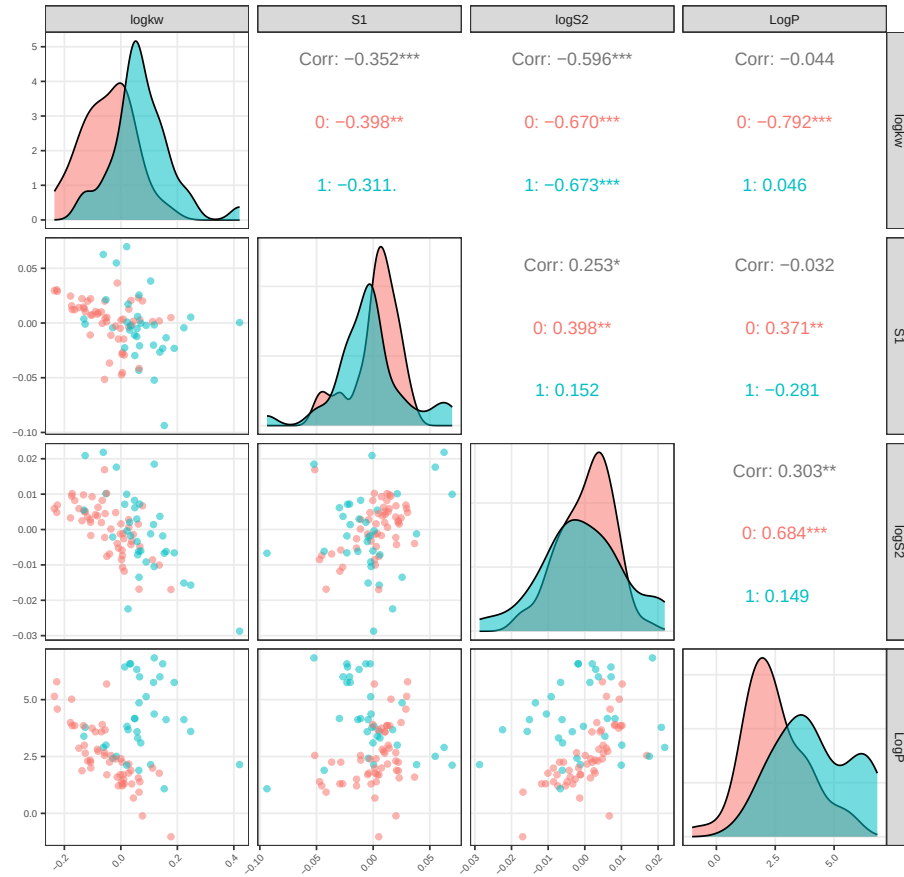
CSH Phenyl-Hexyl



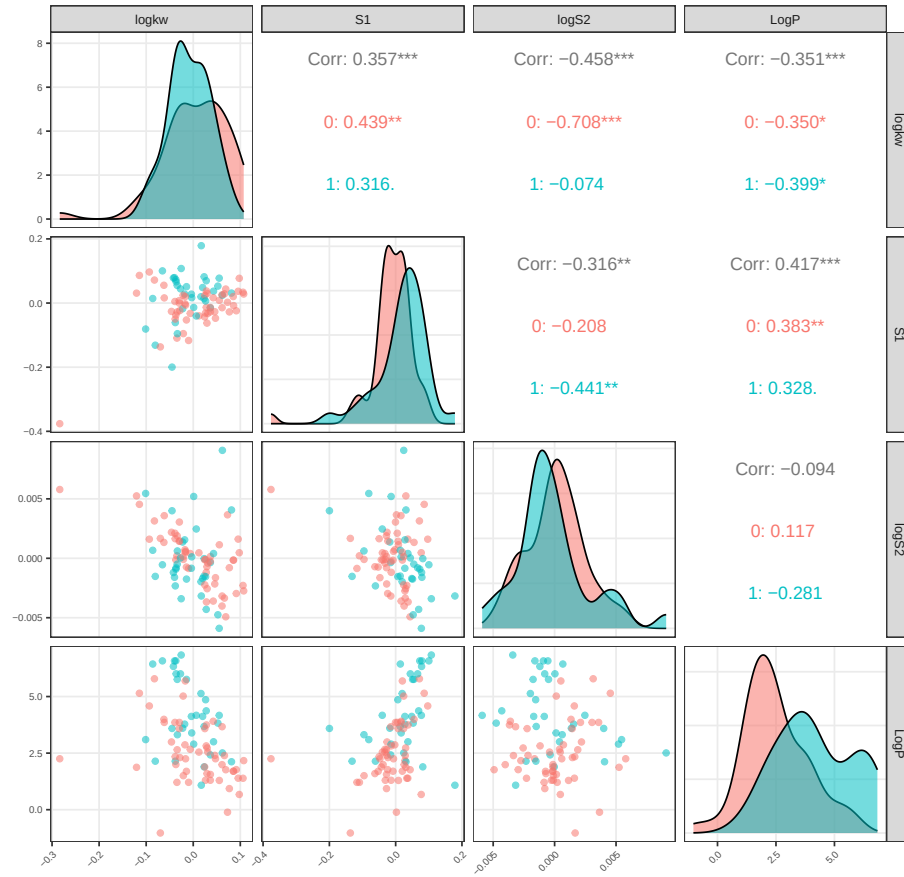
HSS PFP



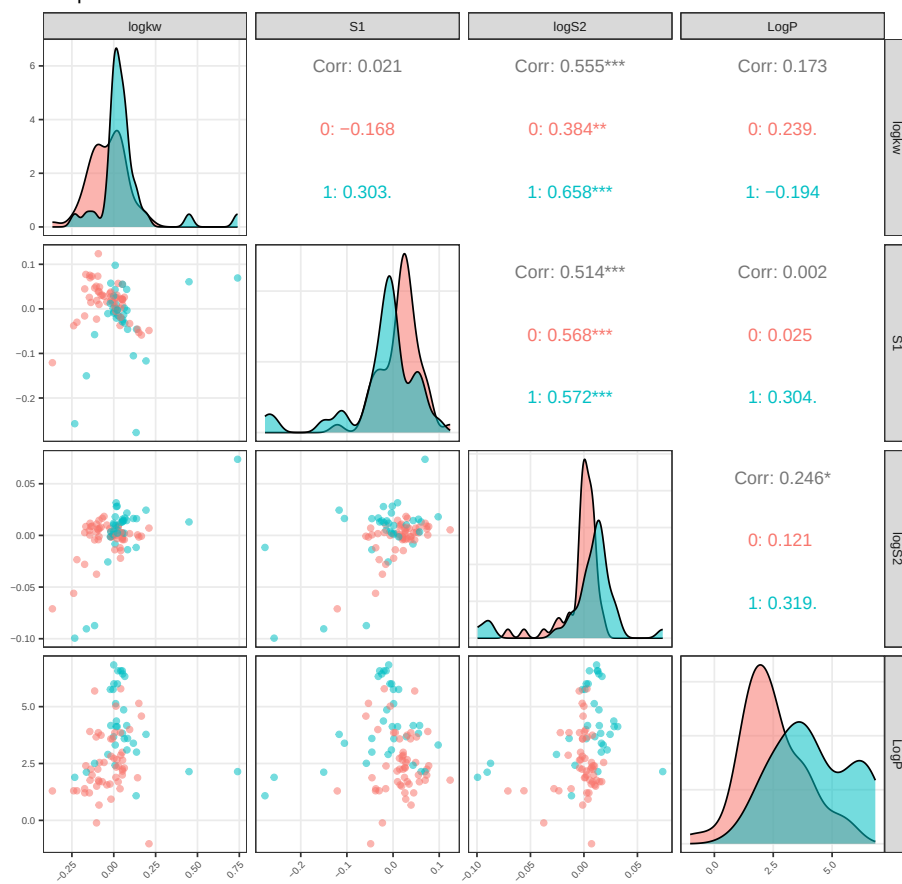
SB-C8



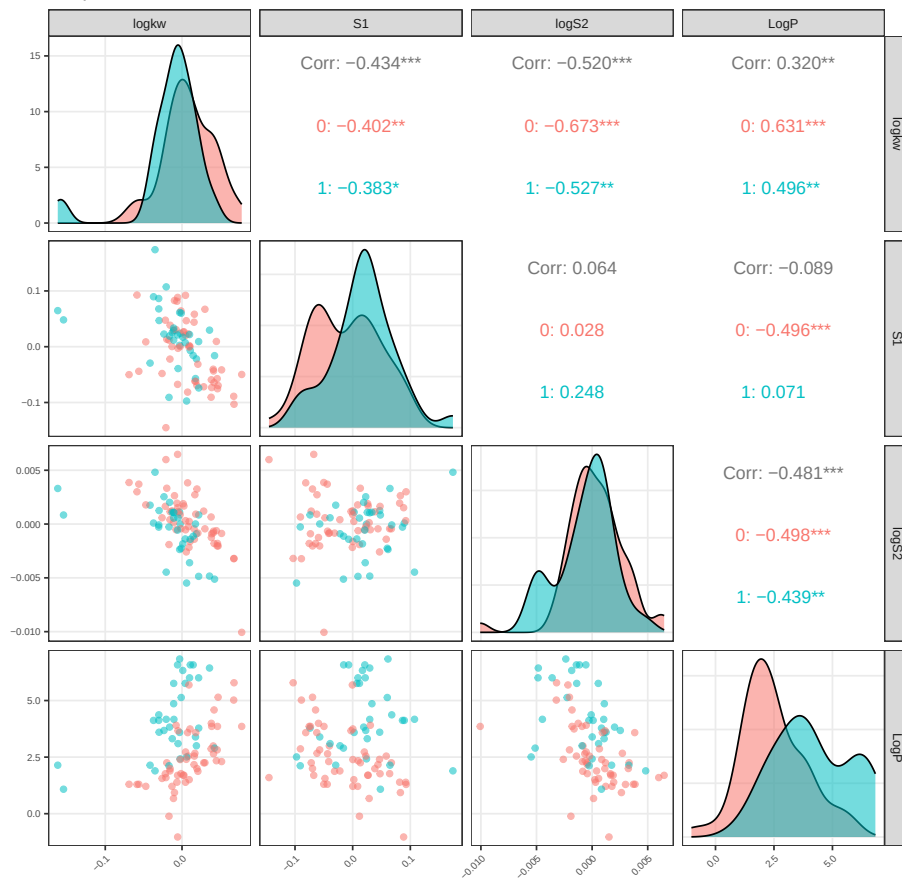
BEH C8



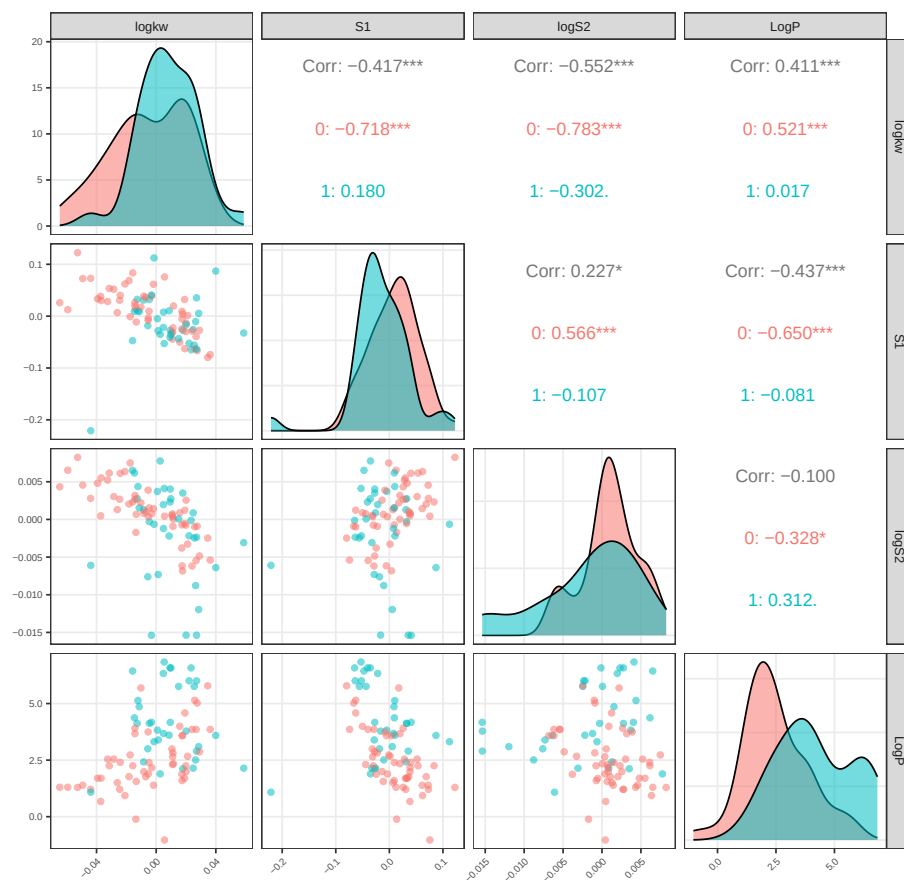
Eclipse PAH



Eclipse Plus C18



SB-C18



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Original Computing Environment

R version 4.3.1 (2023-06-16 ucrt)
Platform: x86_64-w64-mingw32/x64 (64-bit)
Running under: Windows 10 x64 (build 19045)

Matrix products: default

locale:

```
[1] LC_COLLATE=Polish_Poland.utf8 LC_CTYPE=Polish_Poland.utf8
[3] LC_MONETARY=Polish_Poland.utf8 LC_NUMERIC=C
[5] LC_TIME=Polish_Poland.utf8
```

time zone: Europe/Warsaw

tzcode source: internal

attached base packages:

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

other attached packages:

```
[1] kableExtra_1.4.0 dplyr_1.1.4
```

loaded via a namespace (and not attached):

```
[1] vctrs_0.6.5      svglite_2.1.3    cli_3.6.2        knitr_1.46
[5] rlang_1.1.3      xfun_0.54        stringi_1.8.4    generics_0.1.3
[9] textshaping_0.4.0 jsonlite_1.8.8   glue_1.7.0       colorspace_2.1-0
[13] htmltools_0.5.8.1 ragg_1.3.2       scales_1.3.0     fansi_1.0.6
[17] rmarkdown_2.27   munsell_0.5.1    evaluate_1.0.3   tibble_3.2.1
[21] fastmap_1.2.0    yaml_2.3.8       lifecycle_1.0.4  stringr_1.5.1
[25] compiler_4.3.1   codetools_0.2-19 pkgconfig_2.0.3  rstudioapi_0.16.0
[29] systemfonts_1.1.0 digest_0.6.35    viridisLite_0.4.2 R6_2.5.1
[33] tidyselect_1.2.1 utf8_1.2.4       pillar_1.9.0     magrittr_2.0.3
[37] withr_3.0.2      tools_4.3.1      xml2_1.3.6
```